NSE: Non-Destructive Supervision Extraction from Noisy Partial Labels

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Abstract

Noisy Partial Label Learning (NPLL) requires learning from candidate label sets that may contain false positives and, in the noisy case, may even exclude the ground-truth class. Many PLL and NPLL methods obtain stronger supervision by purifying, correcting, reconstructing, or reweighting candidate labels, but model-induced decisions that are carried across epochs can become persistent components of the working supervision. We propose ***NSE*** *(****N****on-Destructive* ***S****upervision* ***E****xtraction), a framework based on* epoch-wise source restoration. *NSE separates the native weak-supervision record from the epoch-local working supervision: at the beginning of each epoch, the working prior is restored from the original candidate mask or crowd prior, and all denoising, disambiguation, topology propagation, pseudo-label extraction, reliable selection, and salvage are applied only to temporary evidence. NSE uses Topology-DAES, a topology-guided Dynamic Adaptive Entropy Similarity kernel, for supervision extraction. Topology-DAES combines masked-entropy topology reliability with entropy-adaptive KNN propagation, allowing consistent neighborhoods to propagate sharper evidence while suppressing ambiguous weak-label neighborhoods. Model predictions are introduced only as temporary semantic evidence: they are gradually warmed up, projected by the native prior, and fused with the Topology-DAES belief through sample-wise reliability arbitration. Reliable selection and consensus-based salvage update training eligibility, not the persistent native record. Across synthetic NPLL, CIFAR-100H, and real crowdsourced datasets, NSE achieves strong performance without cross-epoch source write-back; reset-versus-write-back stress tests further show that persistent source insertion causes large source drift and wrong-write rates. These results suggest that NPLL robustness can be obtained by extracting reliable epoch-local supervision from noisy weak labels, rather than by carrying model-induced source modifications forward.*

1. Introduction

Partial Label Learning (PLL) addresses the challenge of weakly supervised learning where each training instance is annotated with a set of candidate labels, only one of which is the ground truth [1, 2, 23]. While standard PLL assumes the ground truth is strictly contained within the candidate set, real-world scenarios such as web-scale crawling and crowdsourced annotation inevitably introduce noise. This gives rise to the more general and challenging problem of Noisy Partial Label Learning (NPLL) [16], where the true label may be absent from the candidate set, rendering conventional disambiguation assumptions invalid.

The candidate set is not necessarily random noise. Instance-dependent PLL shows that incorrect candidate labels are often related to the instance rather than sampled uniformly. VALEN models such feature-dependent candidate labels through latent label distributions [20], while IDGP decomposes the annotation process into the generation of the true label followed by feature-related incorrect candidates [8]. POP further updates the model and progressively purifies each candidate set in instance-dependent PLL [21]. These studies indicate that the observed candidate set can contain structured weak-supervision information induced by the instance and the annotation process. In NPLL and crowdsourced annotation, however, this information is also noisy and may miss the ground-truth class. A robust method should therefore exploit the observed prior without assuming that it is always correct.

To mitigate the impact of label noise, recent approaches have integrated contrastive learning [15, 16], sample separation [5, 13], candidate-set refinement [9], candidate-label reconstruction [7], or label-importance adjustment [19] into the PLL framework. A representative graph-based method, PALS [11], leverages weighted nearest-neighbor pseudo-labelling and label smoothing to construct reliable image–label pairs. These methods demonstrate that geometry, model prediction, and the original candidate set are all useful sources of evidence. They also reveal a design tension: training supervision is often made increasingly model-dependent through purification, refinement, correction, reconstruction, pseudo-target updating, sample promotion, or

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partial-label augmentation. We call any supervision object inherited by later epochs a *persistent supervision state*; this includes candidate membership, reconstructed candidate sets, soft pseudo-targets, label-confidence or importance weights, and reliable/unreliable promotion states. Such operations are motivated by the belief that reducing candidate noise should make the supervision cleaner. However, when model-induced decisions are carried across epochs as persistent supervision state, early prediction errors can remain in the working supervision used later in training.

We challenge the necessity of this persistent carry-over route. The observed candidate set is noisy, but it is not supervision-free. It contains structured information induced by instance ambiguity, annotator uncertainty, and crowd disagreement. Instead of asking only how to purify the candidate set or how to inherit a corrected supervision state, we ask how to extract reliable supervision from the native weak prior without writing epoch-local model decisions back into any later-epoch supervision state. This work argues that NPLL should not be viewed only as a problem of candidate-set correction: high-performance NPLL can be obtained by restoring the native weak-supervision record at each epoch and extracting low-error active supervision from the restored prior.

In this work, we propose **NSE** (**N**on-Destructive **S**upervision **E**xtraction), a framework designed around a reset-before-extraction principle. NSE does not claim that weak supervision should never be transformed. Instead, it separates the native weak-supervision record from epoch-local working supervision and from model-side learning state. At the beginning of each epoch, the working prior is restored from the original candidate/crowd prior; Topology-DAES propagation, model-evidence projection, graph-semantic fusion, reliable selection, salvage, and pseudo-label training then operate on temporary evidence that is discarded after the epoch. The term non-destructive is therefore restricted to persistent supervision state: model parameters, optimizer state, prototypes, and diagnostic histories are still learned across epochs, but model-induced candidate membership, pseudo-targets, or sample-promotion decisions are not inherited as the supervision source for later epochs. NSE addresses supervision extraction with **Topology-DAES**, a topology-guided Dynamic Adaptive Entropy Similarity kernel. Consistent neighborhoods provide sharp geometric evidence, while ambiguous weak-label neighborhoods are prevented from aggressively spreading unreliable labels. Model predictions are used only as temporary semantic evidence and must pass through the native prior before entering the second Topology-DAES propagation. Training is then performed on an active set consisting of reliable pseudo-labels and samples recovered by stable model–KNN–prototype consensus, while ordinary non-reliable samples are assigned zero sampling weight.

Our contributions are summarized as follows:

- **We identify a persistent supervision-state risk in NPLL.** Repeatedly purifying, correcting, reconstructing, promoting, or reweighting weak labels can make later-epoch supervision model-dependent and may turn early biased predictions into inherited supervision state.
- **We propose epoch-wise source restoration.** NSE restores the working prior from the native candidate/crowd record at the beginning of each epoch and focuses on extracting reliable active supervision through graph-semantic evidence fusion.
- **We introduce Topology-DAES as the supervision extraction kernel of NSE.** It uses masked-entropy topology reliability to measure local agreement between neighbor votes and the native weak prior, and uses entropy-adaptive temperature to control propagation under ambiguity.
- **We combine Topology-DAES geometric evidence with prior-constrained model evidence.** Model predictions provide temporary semantic evidence and must be projected by the native prior before the second propagation stage.
- **We introduce active-set training with consensus-based salvage.** Reliable selection and salvage update training eligibility, not candidate membership.
- **We empirically challenge the necessity of persistent supervision-state carry-over.** Across synthetic, hierarchical, and crowdsourced NPLL settings, and through prior-aligned persistent-state proxy tests, high performance can be achieved without inheriting model-induced supervision-state modifications across epochs.

2. Related Work

Our work is situated at the intersection of weakly supervised learning, label noise robustness, and graph-based evidence aggregation. We group prior work by the way it uses the observed candidate labels.

2.1. Standard and Instance-Dependent PLL

Partial Label Learning (PLL) centers on label disambiguation: identifying the unique ground-truth label concealed within a set of candidate labels. Early work formalized learning from ambiguously labeled images, where each instance is associated with a candidate label set containing one correct label [1]. Cour et al. introduced the ambiguity degree to analyze when ambiguous supervision can control true classification error up to an ambiguity-dependent factor, and proposed a convex surrogate loss for ambiguous multiclass classification. Later PLL studies explored instance-based propagation [23], maximum-margin formulations [22], risk-consistent objectives [2], and weighted losses [17]. PRODEN progressively identifies true labels by jointly updating the model and the candidate-supported label confidence [6]. Deep PLL methods further combine representation learning with disambiguation, including CRDPLL-style consistency regularization [18]. CroSel uses a FIFO memory bank of historical model predictions and cross selection between two networks to select confident candidate-supported pseudo-labels [14]. These methods usually assume that the true label is contained in the candidate set. Many standard PLL losses keep the observed candidate membership fixed and use it through the loss, risk estimator, label weights, or current soft targets rather than rewriting the candidate source. This early perspective is important for our work: the observed candidate set is not supervision-free noise, but a weak supervisory object whose usefulness depends on the ambiguity structure. NSE extends this view to noisy partial labels by extracting epoch-local active supervision from the restored native prior, rather than persistently rewriting the candidate source.

Instance-dependent PLL relaxes the random-noise view of candidate generation. VALEN associates each instance with a latent label distribution and recovers the distribution through variational label enhancement [20]. IDGP further decomposes instance-dependent candidate generation into two sequential parts: the correct label is generated first, and feature-related incorrect labels are then selected because of annotation uncertainty [8]. POP progressively updates the learning model and purifies candidate label sets, with the goal of expanding the reliable region [21]. These works show that candidate quality is sample-dependent and that the observed candidate set can encode structured annotation information. Their main operation is still candidate-set disambiguation or purification, whereas our method uses this observation to motivate preserving the native candidate/crowd prior as a constraint on later evidence fusion.

2.2. Noisy PLL and Persistent Supervision State

Noisy Partial Label Learning (NPLL) is more difficult because the true label may be absent from the observed candidate set. PiCO [15] and PiCO+ [16] use prototype-based contrastive learning and robust sample selection to improve deep PLL under ambiguity and noise. IRNet [5] detects noisy samples and performs label correction. UPLLRS [13] recursively separates reliable and unreliable samples. FREDIS [9] explicitly combines disambiguation and refinement by moving incorrect labels out of the candidate set and moving predicted labels into it. Recent NPLL reconstruction work also separates noisy samples and reconstructs candidate labels when the true label is missing [7]. ALIM [19] reduces the impact of detection errors by adjusting the importance of the initial candidate set and model outputs.

These methods provide strong baselines, but they expose a design tension that cannot be reduced to a single "source rewrite" category. Standard PLL usually assumes the true label is already inside the candidate set; many methods therefore disambiguate a fixed PL source through losses, risk estimators, label weights, or current candidate-supported soft targets. NPLL breaks this assumption because the true label may be absent, so recent methods more often need correction, refinement, promotion, augmentation, or pseudo-target recovery. Existing methods differ in where model-induced supervision is stored. Some methods preserve candidate membership and modify only the loss, risk estimate, or current label-importance weights; ALIM is a close NPLL example because it trades off the initial candidate set and model outputs without a hard candidate rewrite. Some methods maintain soft pseudo-target, label-confidence, or label-distribution states, such as PRODEN, LWS, PiCO, and VALEN. Others modify or reconstruct candidate membership, as in POP, FREDIS, IRNet, NPLL reconstruction, and partial-label augmentation in PALS-style training. UPLLRS instead carries reliable/unreliable sample-state information. Thus, the relevant distinction is not whether a method uses model-induced supervision; most strong methods do. The key question is whether that supervision is carried into later epochs as a *persistent supervision state*: candidate membership, reconstructed candidate sets, pseudo-targets, label-importance weights, or sample-reliability state. When noisy-sample detection or pseudo-label prediction is wrong, such carry-over can preserve the error through later training. This motivates an alternative: restore the working prior from the native weak-supervision record at the beginning of each epoch and use model predictions only as temporary extraction evidence.

2.3. Graph-Based Extraction from Weak Supervision

Graph-based PLL uses the local feature neighborhood as an additional source of weak supervision. IPAL [23] performs instance-based label propagation in standard PLL. PALS [11] extends the KNN perspective to NPLL by exploiting noisy partial labels through weighted nearest-neighbor pseudo-labelling and label smoothing. These methods indicate that local geometry can recover useful supervision even when individual candidate sets are ambiguous.

NSE follows this graph-based direction but changes the role of the KNN graph. Instead of using a fixed similarity kernel to generate pseudo-labels, NSE uses Topology-DAES as a supervision extraction kernel. Topology-DAES first evaluates whether local neighbor votes are consistent with the native weak prior through masked-entropy reliability, and then adapts the propagation temperature accord-

Method	Supervision state	Representative update	Carry-over	Potential limitation under carry-over	NSE distinction
PRODEN / LWS [6, 17]	label-confidence / label-weight state	update candidate-supported confidence or weights while keeping membership fixed	soft	mistaken confidence concentration can bias soft targets, but does not hard-edit candidate membership	restores prior and extracts topology-controlled active evidence
CroSel [14]	history queue / selected pseudo-label set	FIFO memory bank supports cross-network selection of stable, confident candidate labels	selected-set state	selected labels depend on historical prediction stability under the standard PLL assumption that the true label is in the candidate set	queue stability inspires salvage, but NSE uses tri-consensus and restores the native NPLL prior
POP [21]	candidate membership	progressive purification removes low-confidence labels	yes	mistaken deletion may remove useful ambiguous support from later training	no persistent candidate deletion
FREDIS [9]	candidate membership	score-gap refinement inserts non-candidates and score-gap disambiguation removes candidates	yes	later disambiguation can remove mistaken refinement labels, but both procedures update the working candidate-set state across fusion rounds	restore native prior before extraction
NPLL reconstruction [7]	reconstructed candidate set	sample separation and KNN/model pseudo-label evidence reconstruct shorter candidate label sets for training	working reconstruction	training optimizes reconstructed membership rather than an extracted active set anchored by restored prior	extract active supervision, not a new candidate state
PALS [11]	pseudo-label/partial-label augmentation	confident top-1 predictions can augment the next-iteration partial label	partial	model-updated features and predictions can influence subsequent pseudo-labelling and partial-label augmentation	model evidence remains epoch-local and prior-constrained
UPLLRS [13]	split/training-stage state	recursively separates reliable and unreliable subsets for disambiguation and semi-supervised training	yes	errors in the reliable/unreliable split can affect subsequent disambiguation and semi-supervised training	active eligibility is regenerated each epoch
IRNet [5]	correction state	candidate/non-candidate score-gap detection followed by non-candidate insertion	yes	correction quality depends on the chosen detection stage and is then carried forward	no correction state is written back
PiCO/PiCO+ [15, 16]	pseudo-target state	pseudo targets are updated and influence contrastive positive pairs	yes	pseudo-target errors may affect later target construction	pseudo-labels are temporary active targets
ALIM [19]	importance weights / adjusted training targets	recomputes weights from initial candidate set and model outputs	no hard rewrite	close to source preservation, but adjustment happens at prediction-level importance rather than topology-controlled extraction	topology extraction from restored prior
NSE	restored working prior	$\tilde{\Omega}_i^t \leftarrow \Omega_i^0$, then temporary evidence extraction	**no**	depends on extraction quality and active-set precision	**epoch-wise source restoration**

Table 1. Persistent supervision state versus epoch-wise restored extraction in PLL/NPLL. The comparison separates where model-induced supervision is stored: some methods keep candidate membership fixed but update weights or targets, while others carry candidate membership, reconstructed candidate sets, pseudo-targets, label-importance weights, or sample-reliability states across epochs. NSE distinguishes the native source record Ω_i^0 from temporary working evidence and restores the working prior before each extraction stage.

ing to neighborhood entropy. Thus, the KNN graph is not merely a pseudo-label generator; it becomes a reliability-controlled extractor from an epoch-wise restored weak prior. Instead of replacing or rewriting the candidate set, the native candidate/crowd prior constrains the model belief before it enters the second propagation stage. The reliability ratio r_i performs sample-wise arbitration between Topology-DAES geometric evidence and temporary model evidence. PiCO+ further shows that distance to class prototypes can help distinguish clean and noisy examples under NPLL. NSE uses the same intuition in a weaker, non-persistent form: class-center scores participate in the model–KNN–prototype consensus used to salvage active samples, but the prototype-induced pseudo-target is not written into a later-epoch supervision state. Reliable selection and salvage affect training eligibility, not the persistent native candidate/crowd record.

3. Method

3.1. Problem Setup

We consider Noisy Partial Label Learning (NPLL). The training set is $\mathcal{D} = \{(\mathbf{x}_i, Y_i)\}_{i=1}^N$, where $Y_i \subseteq \{1, \dots, C\}$ is a candidate label set. In standard PLL the ground-truth label is assumed to be contained in Y_i, whereas in NPLL this assumption may be violated by noisy annotations. For real crowdsourced datasets, Y_i is replaced by a soft crowd prior that summarizes multiple annotators. The goal is to train an encoder f_θ and classifier h_ϕ while avoiding cross-epoch carry-over of model-induced supervision state.

3.2. Overview of NSE

The proposed method is **NSE**. It replaces persistent supervision-state refinement with an epoch-wise restored supervision extraction path. Each epoch contains four steps:

- build a single-head cosine KNN graph from weak-view features and run the first Topology-DAES extrac-

Algorithm 1 Epoch-wise Source Restoration in NSE. Steps 1–7 are repeated every epoch; only model-side learning state is carried across epochs, while candidate membership, pseudo-targets, and sample-promotion state are regenerated as temporary evidence.

1. Restore working prior: $\tilde{\Omega}_i^t \leftarrow \Omega_i^0$ for all i.
2. Extract weak-view features and build a cosine KNN graph G^t.
3. Run first Topology-DAES on $\tilde{\Omega}^t$ to obtain P_{geo}^t.
4. Compute weak/strong model belief and warm it up into $P_{\text{model}}^{\text{eff},t}$.
5. Project model evidence by $\tilde{\Omega}^t$ and fuse it with P_{geo}^t.
6. Run second Topology-DAES to obtain $P^{(2),t}$ and pseudo-labels.
7. Select reliable set R_t and salvage set S_t; set $A_t = R_t \cup S_t$.
8. Update model parameters with active-set MixUp training.
9. Update model-side states such as prototypes and histories; discard epoch-local evidence.

tion to obtain geometric evidence P_{geo} from the restored working prior;

- compute weak/strong model predictions and introduce them gradually through warm-up;
- project model evidence by the native candidate/crowd prior and fuse it with P_{geo} through sample-wise reliability arbitration;
- run the second Topology-DAES propagation and train on active samples formed by reliable pseudo-labels and stable consensus-based salvaged samples.

The central fusion rule is

$$P_{\text{in},i}^{(2)} = \text{Norm}\Big(r_i P_{\text{geo},i} + (1 - r_i)\hat{P}_{\text{model},i}\Big), \quad (1)$$

where $\hat{P}_{\text{model},i}$ is constrained by the restored native prior. Thus the model can influence propagation within the epoch, but it cannot write its prediction into the next epoch's source state.

3.3. Epoch-Wise Source Restoration Principle

NSE does not claim that weak supervision should never be transformed. It separates the native weak-supervision record from epoch-local working supervision and from later-epoch persistent supervision state. We distinguish four kinds of state. The *native source record* Ω_i^0 is the stored candidate mask for synthetic NPLL data or the stored soft crowd prior for real crowdsourced data. The *epoch-local evidence* includes P_{geo}^t, $P_{\text{in}}^{(2),t}$, $P^{(2),t}$, pseudo-labels, and active eligibility. The *persistent supervision state* is any supervision object inherited by later epochs, including candidate membership, reconstructed candidate sets, soft pseudo-targets, label-confidence or importance weights, and reliable/unreliable promotion states. The *model-side learning state* includes network weights, optimizer state, prototype exponential moving averages, and temporal diagnostic histories. NSE is non-destructive with respect to persistent supervision state, not with respect to model learning state. At the beginning of epoch t, NSE restores the working prior from the native source:

$$\tilde{\Omega}_i^t = \text{Restore}(\Omega_i^0) = \Omega_i^0. \quad (2)$$

All denoising, disambiguation, topology propagation, model projection, pseudo-label extraction, reliable selection, and salvage are then performed on temporary evidence:

$$E_i^t = \text{Extract}\left(\tilde{\Omega}_i^t, f_\theta^{t-1}, G^t\right), \qquad A_t = \text{Select}(E^t), \quad (3)$$

where E_i^t includes quantities such as $P_{\text{geo},i}^t$, $P_{\text{in},i}^{(2),t}$, $P_i^{(2),t}$, and the current pseudo-labels used for active training. These quantities can be denoised, sharpened, filtered, or expanded through consensus within the epoch, but they are not written back into candidate membership, pseudo-target, label-weight, or sample-promotion state for the next epoch.

Instead, NSE learns an extraction path:

$$\Omega_i^0 \to \tilde{\Omega}_i^t \to P_{\text{geo},i}^t \to P_{\text{in},i}^{(2),t} \to A_t, \quad (4)$$

where $P_{\text{geo},i}^t$ and $P_{\text{in},i}^{(2),t}$ are temporary evidence distributions and A_t is the active training set. After the epoch, the working evidence is discarded and the next epoch again starts from Ω_i^0.

This principle separates NSE from persistent carry-over updates. A persistent supervision-state update can be abstracted as

$$S_i^{t+1} = U\left(S_i^t, E_i^t; f_\theta^t\right), \quad (5)$$

where S_i^t denotes whatever supervision state will be reused by the next epoch. It may be a candidate set, a reconstructed candidate set, a soft pseudo-target, a label-importance vector, or a sample-reliability/promotion state. For a soft write-back form,

$$S_i^{t+1} = \text{Norm}\big((1 - \alpha_{\text{wb}})S_i^t + \alpha_{\text{wb}} E_i^t\big), \quad (6)$$

an update error rate e_t can accumulate recursively as

$$\rho_{t+1} \le (1 - \alpha_{\text{wb}})\rho_t + \alpha_{\text{wb}} e_t. \quad (7)$$

NSE prevents this persistent supervision-state drift channel by restoring $\tilde{\Omega}_i^{t+1}$ from Ω_i^0 before the next extraction stage and by regenerating active eligibility from epoch-local evidence. Model errors can still affect temporary evidence and active-set decisions in the current epoch, but they are not carried as persistent candidate membership, pseudo-target, or sample-promotion modifications.

3.4. Dual-View Model Prediction

For each sample, the data loader provides a weak and a strong augmented view. The weak view is used to construct the KNN feature graph because it preserves a cleaner local topology:

$$\mathbf{z}_i = \frac{f_\theta(\mathbf{x}_i^w)}{\|f_\theta(\mathbf{x}_i^w)\|_2}. \tag{8}$$

The model belief is computed from both views and averaged:

$$P_{\mathrm{model},i} = \frac{1}{2}\left[\sigma(h_\phi(f_\theta(\mathbf{x}_i^w))) + \sigma(h_\phi(f_\theta(\mathbf{x}_i^s)))\right]. \tag{9}$$

This "separate model" design makes the semantic signal less dependent on one augmentation path while keeping the geometric graph tied to the weak-view feature manifold.

3.5. Topology-DAES Supervision Extraction

Given normalized features $\mathbf{z}_i$, NSE builds a single-head cosine K-nearest-neighbor graph. Topology-DAES is the supervision extraction kernel of NSE, where DAES denotes Dynamic Adaptive Entropy Similarity. It is not a generic KNN weighting trick: it determines how supervision is extracted from the epoch-wise restored weak prior.

Masked-entropy topology reliability. For a reference distribution P, let S_{ij} be the cosine similarity between sample i and neighbor $j \in \mathcal{N}_K(i)$. Before using a similarity as a linear vote weight, NSE clamps it to the non-negative range, $\bar{S}_{ij} = \max(S_{ij}, 0)$, so the resulting neighborhood vote remains a valid non-negative aggregation. NSE first computes a topology vote

$$\begin{aligned} \alpha_{ij} &= \frac{\bar{S}_{ij}}{\sum_{j' \in \mathcal{N}_K(i)} \bar{S}_{ij'} + \epsilon}, \\ P_i^{\mathrm{top}} &= \mathrm{Norm}\left(\sum_{j \in \mathcal{N}_K(i)} \alpha_{ij} P_j\right). \end{aligned} \tag{10}$$

The topology vote is compared with the self weak prior

$$P_i^{\mathrm{self}} = \mathrm{Norm}(P_i). \tag{11}$$

If M_i is numerically empty, the normalization uses ϵ smoothing and falls back to a high-entropy agreement score. We use masked entropy to measure local agreement:

$$M_i = P_i^{\mathrm{top}} \odot P_i^{\mathrm{self}}, \qquad \tilde{M}_i = \mathrm{Norm}(M_i), \tag{12}$$

$$s_i = -\frac{1}{\log C}\sum_c \tilde{M}_{ic}\log(\tilde{M}_{ic} + \epsilon), \qquad \rho_i = \exp(-\gamma s_i^2). \tag{13}$$

If neighbor votes and the native weak prior concentrate on the same labels, s_i is small and ρ_i is high. If the neighborhood conflicts with the weak prior or spreads mass over many labels, s_i increases and ρ_i decreases. This masked-entropy reliability is the only topology reliability mode used in NSE.

Entropy-adaptive DAES temperature. Topology reliability measures whether the local weak-label topology is trustworthy. NSE further uses DAES to adapt propagation sharpness to neighborhood uncertainty. For a reference distribution P, the spatially weighted neighborhood mean is

$$\bar{P}_i = \frac{\sum_{j \in \mathcal{N}_K(i)} W_{ij}^{\mathrm{sp}} P_{\mathrm{ref},j}}{\sum_{j \in \mathcal{N}_K(i)} W_{ij}^{\mathrm{sp}}}, \qquad W_{ij}^{\mathrm{sp}} = \mathrm{softmax}_j\left(\frac{S_{ij}}{\tau_{\mathrm{sp}}}\right). \tag{14}$$

Its normalized entropy is

$$H_i = -\frac{1}{\log C}\sum_{c=1}^{C} \bar{P}_{i,c}\log(\bar{P}_{i,c} + \epsilon). \tag{15}$$

DAES then forms a sample-adaptive temperature

$$\tau_i = \tau_0 + \lambda_H H_i^2, \tag{16}$$

so low-entropy neighborhoods receive sharper propagation and high-entropy neighborhoods receive flatter propagation.

Topology-DAES propagation. The final Topology-DAES edge score uses the reliability of the neighbor node:

$$\tilde{S}_{ij} = \bar{S}_{ij}\rho_j. \tag{17}$$

Thus, a neighbor with unreliable weak-label topology contributes less evidence even if it is close in feature space. The entropy-adaptive propagation weight is

$$W_{ij}^{\mathrm{td}} = \exp\left(\frac{\tilde{S}_{ij}^{p_{\mathrm{sim}}}}{\tau_i} - \max_{j' \in \mathcal{N}_K(i)} \frac{\tilde{S}_{ij'}^{p_{\mathrm{sim}}}}{\tau_i}\right). \tag{18}$$

We use $p_{\mathrm{sim}} = 2$ in all reported experiments, matching the squared similarity score used by the implementation. The Topology-DAES belief is

$$P_i^{\mathrm{td}} = \mathrm{softmax}\left(\sum_{j \in \mathcal{N}_K(i)} W_{ij}^{\mathrm{td}} P_j\right). \tag{19}$$

The same softmax normalization is used in both propagation passes; the alternative exponential-kernel ablation changes the edge score, not this normalization step.

The first Topology-DAES pass uses the restored candidate/crowd prior as input and produces P_{geo}. The second pass uses the fused graph–semantic evidence $P_{\text{in}}^{(2)}$ as input and produces the final pseudo-label distribution for active-set training.

3.6. Progressive Model Warm-Up

The model branch is introduced gradually to avoid early self-confirmation. At epoch t, its maximum contribution is controlled by

$$w_t = \min\left(w_{\max}, \frac{t}{T_{\text{warm}}}\right), \tag{20}$$

where T_{warm} is the model warm-up length and $w_{\max}$ is the maximum model-evidence cap. The effective model belief is

$$P_{\text{model},i}^{\text{eff}} = w_t P_{\text{model},i} + (1 - w_t) P_{\text{geo},i}. \tag{21}$$

At the beginning of training, the model branch collapses to the graph belief; after warm-up, it contributes semantic evidence up to the specified cap.

3.7. Reliability-Aware Graph–Semantic Fusion

NSE uses a confidence-ratio reliability score. Let

$$s_i^{\text{geo}} = \max_c P_{\text{geo},i,c}, \qquad s_i^{\text{model}} = \max_c P_{\text{model},i,c}. \tag{22}$$

The reliability assigned to the KNN belief is

$$r_i = \frac{s_i^{\text{geo}}}{s_i^{\text{geo}} + s_i^{\text{model}} + \epsilon}. \tag{23}$$

Large r_i means the geometric belief is sharper than the model belief; small r_i allows the model belief to correct ambiguous neighborhoods.

The external prior $\tilde{\Omega}_i^t$ is restored from the native weak-supervision constraint: candidate mask for synthetic NPLL data and crowd prior for real crowdsourced data. The model belief is first projected into this prior:

$$\hat{P}_{\text{model},i} = \text{Norm}\left(P_{\text{model},i}^{\text{eff}} \odot \tilde{\Omega}_i^t\right). \tag{24}$$

The second-stage propagation input is then Eq. (1). This is the unified graph–semantic evidence path used for CIFAR, CIFAR-100H, Benthic, Plankton, and Treeversity.

Algorithmic advantages. This fusion rule provides advantages that are not simultaneously present in existing NPLL designs. In contrast to refinement-based methods such as FREDIS [9], NSE does not carry an add/remove candidate-set state across epochs. In contrast to progressive purification [21], reliable-set expansion in our method changes which samples are used for optimization in the current epoch, not the original candidate membership. In contrast to candidate-label reconstruction [7], NSE does not optimize on a reconstructed candidate set as the new working source; it extracts active supervision from a restored native prior. In contrast to detection-and-correction frameworks such as IRNet [5] and recursive separation [13], the method does not require a correction or promotion state to be carried forward. In contrast to prototype-contrastive disambiguation [15, 16], the model branch is not the sole source of semantic correction; it must compete with the KNN belief through r_i and the original prior through Eq. (24). Compared with ALIM [19], which is closest to source preservation because it trades off the initial candidate set and model outputs, NSE performs topology-controlled extraction from the restored prior and updates only active training eligibility.

3.8. Recovering Candidate-Missing Samples

NPLL differs from standard PLL because some samples have $y_i \notin Y_i^0$. NSE handles these samples by separating the role of the native prior in the model branch from the role of graph and consensus evidence. The prior projection in Eq. (24) constrains only the temporary model evidence: it prevents an early classifier prediction from directly writing an unsupported class into the working source. It does not erase non-candidate evidence already present in P_{geo}, because P_{geo} is obtained by propagating neighbors' restored priors. Thus a candidate-missing sample can receive mass on its true class if nearby samples carry that class in their own native priors.

Reliable-set admission remains conservative: the pseudo-label must be supported by the sample's candidate/crowd prior. Therefore, initially noisy samples whose true class is absent from Y_i^0 are not counted as source-level repairs. They are recovered through the salvage path when model, KNN, and prototype predictions become stable and agree over the temporal window. This is the intended distinction: NSE does not repair the native source membership, but it can still recover missing-label samples as epoch-local active supervision.

3.9. Second Propagation and Reliable Set Selection

The fused distribution $P_{\text{in}}^{(2)}$ becomes the reference for the second Topology-DAES propagation:

$$P_i^{(2)} = \text{softmax}\left(\sum_{j \in \mathcal{N}_K(i)} W_{ij}^{(2)} P_{\text{in},j}^{(2)}\right). \tag{25}$$

The final pseudo-label is

$$\hat{y}_i = \arg\max_c P_{i,c}^{(2)}. \tag{26}$$

A sample is placed into the reliable set if the pseudo-label is supported by the candidate/crowd prior and its confidence passes the quantile threshold controlled by δ. Concretely, within each predicted class we keep the highest-confidence fraction determined by the class-count quantile δ, which avoids letting large classes dominate the reliable set. This preserves the two-pass reliable selection structure while replacing branch-specific refinement with a single restored-prior extraction rule.

3.10. Recoverability Intuition

The KNN stage can provide useful supervision only when the propagated true-label mass is larger than the strongest ambiguous-label mass. Let

$$S_i(c) = \sum_{j \in \mathcal{N}_K(i)} w_{ij} P_j^{(0)}(c), \qquad \sum_j w_{ij} = 1, \quad w_{ij} \geq 0 \tag{27}$$

be the neighborhood vote for class c before normalization. If the true label of sample i is y_i, define the KNN supervision margin

$$\Delta_i = S_i(y_i) - \max_{c \neq y_i} S_i(c). \tag{28}$$

A KNN pseudo-label is correct whenever $\Delta_i > 0$. In NPLL, noisy candidate sets reduce $S_i(y_i)$ when the true label is absent and increase the competing ambiguous-label scores. Therefore, the central question is whether neighbor aggregation restores a positive margin.

This margin also gives an intuition for reliable supervision. The following calculation is not a full guarantee for the adaptive training dynamics; it is made under simplifying assumptions, conditioned on the learned representation and graph, and treats neighbor votes as bounded random variables. For any competing class $c \neq y_i$, let

$$D_{i,c} = \sum_{j \in \mathcal{N}_K(i)} w_{ij} \left[P_j^{(0)}(y_i) - P_j^{(0)}(c) \right], \qquad m_{i,c} = \mathbb{E}[D_{i,c}]. \tag{29}$$

Since each summand is bounded in $[-w_{ij}, w_{ij}]$, Hoeffding's inequality gives the illustrative bound

$$\Pr(D_{i,c} \leq 0) \leq \exp\left(-\frac{m_{i,c}^2}{2 \sum_j w_{ij}^2} \right). \tag{30}$$

By a union bound over all wrong classes,

$$\Pr(\hat{y}_i^{\text{knn}} \neq y_i) \leq \sum_{c \neq y_i} \exp\left(-\frac{m_{i,c}^2}{2 \sum_j w_{ij}^2} \right). \tag{31}$$

For uniform neighbors, $\sum_j w_{ij}^2 = 1/K$, so the error bound decreases with larger neighbor count and larger true-label margin. For adaptive weights, the term $(\sum_j w_{ij}^2)^{-1}$ acts as an effective neighbor count. Topology-DAES is therefore useful when it increases the effective margin by sharpening consistent neighborhoods and avoiding over-confident propagation in ambiguous neighborhoods.

Similarly, if the active training set has pseudo-label error rate ε_{act}, then a bounded loss $\ell \in [0, B]$ satisfies the standard-style decomposition

$$R(h) \leq \hat{R}_{\text{act}}(h) + 2\mathfrak{R}_{n_{\text{act}}}(\mathcal{H}) + B\sqrt{\frac{\log(1/\delta_{\text{conf}})}{2n_{\text{act}}}} + B\varepsilon_{\text{act}}, \tag{32}$$

with probability at least $1 - \delta_{\text{conf}}$. Eq. (31) connects the KNN propagation margin to ε_{act}, and $\mathfrak{R}_{n_{\text{act}}}(\mathcal{H})$ denotes the empirical Rademacher complexity of the hypothesis class on the active set. This standard-style expression is used only as intuition: better neighbor aggregation reduces pseudo-label errors in the active set, which reduces the noise term in the downstream decomposition.

This bound also clarifies the design philosophy. Improving NPLL performance does not require cross-epoch inheritance of model-induced supervision state; it requires reducing the error rate of the extracted active supervision ε_{act}. Persistent candidate-set or pseudo-target updates attempt to reduce ε_{act} by carrying model-induced supervision states forward, while NSE reduces it by improving the extraction path from the restored native source Ω_i^0.

3.11. Prototype and Consensus Salvage

NSE maintains class prototypes with exponential moving average. If $\mathcal{R}_{t,c}$ is the reliable subset assigned to class c at epoch t, the class prototype is updated as

$$\mu_c^t = \beta \mu_c^{t-1} + (1 - \beta) \frac{1}{|\mathcal{R}_{t,c}|} \sum_{i \in \mathcal{R}_{t,c}} \mathbf{z}_i, \tag{33}$$

when $\mathcal{R}_{t,c}$ is non-empty, followed by ℓ_2 normalization. Prototype predictions are compared with model and KNN predictions over the unreliable pool:

$$\hat{y}_i^{\text{model}}, \quad \hat{y}_i^{\text{knn}}, \quad \hat{y}_i^{\text{proto}}. \tag{34}$$

Let L_{hist} be the history length and $\mathbf{a}_{i,\tau} = (\hat{y}_{i,\tau}^m, \hat{y}_{i,\tau}^k, \hat{y}_{i,\tau}^p)$ denote the model, KNN, and prototype hard predictions. A sample enters the salvage set if it is not in the reliable set for all epochs in the window and the three predictions are valid, stable, and equal throughout the window:

$$i \in S_t \iff \begin{cases} i \notin R_\tau, \\ \mathbf{a}_{i,\tau} = (\hat{y}_i^S, \hat{y}_i^S, \hat{y}_i^S), \end{cases} \qquad \forall \tau \in [t - L_{\text{hist}} + 1, t]. \tag{35}$$

This temporal queue is inspired by CroSel's memory-bank selection idea, which uses FIFO historical predictions to identify stable candidate-supported pseudo-labels in standard PLL [14]. NSE adapts this idea to NPLL by requiring model–KNN–prototype agreement and by using the selected samples only as epoch-local active supervision, without writing the salvaged label into the native candidate/crowd source. EMA teacher predictions are not used in this consensus rule.

3.12. Active-Set Training Objective

Training uses one unified data loader, but each epoch assigns nonzero sampling weight only to active samples. The active set is the union of the reliable set selected by the second propagation and a salvage set recovered by stable model–KNN–prototype consensus. Samples that are neither reliable nor salvaged are retained in the index space for bookkeeping, but their sampling weight is zero and they do not contribute to the training loss.

The salvage set is therefore a training-set decision rather than a candidate-set update. A salvaged sample can provide supervision in the current epoch when model, KNN, and prototype predictions agree persistently, but its native candidate mask or crowd prior is restored again before the next epoch's extraction. This differs from candidate purification and reconstruction strategies that carry add/remove or reconstructed candidate states forward. The design preserves the native source record and avoids persistent sample-promotion state while still allowing recoverable samples to contribute to optimization through epoch-local active supervision.

Active samples use weak and strong augmented MixUp supervision. For an active sample with pseudo-label $\hat{y}_i$, the smoothed target is

$$\tilde{\mathbf{y}}_i = (1 - \epsilon_{\text{ls}})\mathbf{e}_{\hat{y}_i} + \frac{\epsilon_{\text{ls}}}{C}\mathbf{1}. \tag{36}$$

MixUp draws $\lambda \sim \text{Beta}(\alpha_{\text{mix}}, \alpha_{\text{mix}})$ and optimizes

$$\mathcal{L}_{\text{sup}} = \frac{1}{2}\left[\text{CE}(\tilde{\mathbf{x}}^w, \tilde{\mathbf{y}}) + \text{CE}(\tilde{\mathbf{x}}^s, \tilde{\mathbf{y}})\right]. \tag{37}$$

The total objective used by the current experiments is therefore

$$\mathcal{L} = \mathcal{L}_{\text{active}}, \tag{38}$$

where $\mathcal{L}_{\text{active}}$ has the same weak/strong MixUp form as $\mathcal{L}_{\text{sup}}$ above. This objective is intentionally conservative: ordinary non-reliable samples are not forced into the loss. They can affect training only after they become reliable or are recovered by stable model–KNN–prototype consensus.

3.13. Optimization

The implementation optimizes the encoder and classifier jointly using SGD with momentum. CIFAR-style experiments use ResNet-18 and cosine scheduling by default, while crowdsourced datasets use ResNet-50 with step decay and ImageNet-pretrained initialization. Unless otherwise specified, we use label smoothing $\epsilon_{\text{ls}} = 0$, history length $L_{\text{hist}} = 15$, and Topology-DAES in both propagation passes.

4. Experiments

4.1. Experimental Setup

Datasets and protocols. We evaluate NSE on CIFAR-10 and CIFAR-100 [4], the hierarchical CIFAR-100H setting used in PALS [11], and three real-world crowdsourced datasets: Treeversity, Benthic, and Plankton [12]. For synthetic PLL/NPLL, we follow the standard protocol of PiCO+ [16]. Each incorrect class is added to the candidate set according to the partial rate q, and the ground-truth class is removed with probability η to simulate noisy partial labels. We ensure every training sample has a non-empty observed candidate set. For crowdsourced datasets, the native source Ω_i^0 is a soft crowd prior obtained by normalizing annotator votes; LPI denotes the number of labels per image used to form the prior.

Baselines and metrics. We compare against PLL and NPLL baselines including RC [2], LWS [17], PiCO [15], CRDPLL [18], PiCO+ [16], FREDIS [9], IRNet [5], UPLLRS [13], ALIM [19], and PALS [11]. We report final-epoch test accuracy because NPLL does not assume a clean validation set for model selection. Unless otherwise noted, NSE results are mean and standard deviation over three independent seeds. For baseline methods, the reported numbers are taken from the PALS paper and the PALS-reported comparison tables under the same benchmark protocol. We cite the original baseline papers to identify the compared methods, not to indicate that we independently reran every baseline or copied numbers from the original baseline papers. Entries absent from the PALS comparison are kept unreported. Our NSE runs use the same benchmark backbone family, training budget, augmentation protocol, and validation-free final-epoch test selection used by the PALS protocol.

Diagnostics. For source-restoration analysis, let A_t be the final-epoch active set, $\hat{y}_i^t$ its pseudo-label, Ω_i^0 the native source, and $\tilde{\Omega}_i^t$ the working source. Let $\mathcal{C}_0 = \{i : y_i \in \text{supp}_\epsilon(\Omega_i^0)\}$ be the initially clean samples and $\mathcal{N}_0 = \{i : y_i \notin \text{supp}_\epsilon(\Omega_i^0)\}$ the initially noisy samples. We report

$$\begin{aligned}
\text{Prec}(A_t) &= \frac{1}{|A_t|}\sum_{i \in A_t} \mathbf{1}[\hat{y}_i^t = y_i], \\
\text{Cov}(A_t) &= \frac{|A_t|}{N}, \\
\text{NRR-A}_t &= \frac{1}{|\mathcal{N}_0|}\sum_{i \in A_t \cap \mathcal{N}_0} \mathbf{1}[\hat{y}_i^t = y_i], \\
\text{Drift}_t &= \frac{1}{N}\sum_i \|\tilde{\Omega}_i^t - \Omega_i^0\|_1.
\end{aligned} \tag{39}$$

For persistent source-state variants, W_t is the set of samples whose source row is updated at epoch t. We define $\text{Write}_t = |W_t|/N$, $\text{Wrong}_t = |W_t|^{-1}\sum_{i \in W_t} \mathbf{1}[\hat{y}_i^t \neq y_i]$ when $|W_t| > 0$, $\text{SrcRec-N}_t = |\mathcal{N}_0|^{-1}\sum_{i \in \mathcal{N}_0} \mathbf{1}[y_i \in$

$\mathrm{supp}_\epsilon(\tilde{\Omega}_i^t)]$, and $\mathrm{Damage}_t = 1 - |\mathcal{C}_0|^{-1}\sum_{i\in\mathcal{C}_0}\mathbf{1}[y_i \in \mathrm{supp}_\epsilon(\tilde{\Omega}_i^t)]$. For soft persistent states, we additionally report the true-label mass recovered on initially noisy samples and the true-label mass weakened on initially clean samples:

$$\mathrm{MassRec\text{-}N}_t = |\mathcal{N}_0|^{-1}\sum_{i\in\mathcal{N}_0}\tilde{\Omega}_{i,y_i}^t,$$
$$\mathrm{DamageMass}_t = |\mathcal{C}_0|^{-1}\sum_{i\in\mathcal{C}_0}\max(0, \Omega_{i,y_i}^0 - \tilde{\Omega}_{i,y_i}^t).$$

For persistent sample-promotion variants, W_t is replaced by the set of newly promoted samples, and the same event-error definition is reported as $\mathrm{WrongPromote}_t$. When $|W_t| = 0$, the error rate is not applicable and is reported as "–". We also report AUC-Drift, the mean epoch-wise drift over training, to capture long-term source movement rather than only the final source state.

Implementation details. CIFAR experiments use ResNet-18 and 500 training epochs with SGD, momentum 0.9, weight decay 10^{-3}, batch size 256, and cosine learning-rate decay. For Treeversity, Benthic, and Plankton, we align with the reported PALS setup where specified by using the same ResNet-50 architecture, ImageNet pretrained initialization [3, 10], and 100 training epochs. Since the PALS paper does not fully specify all crowdsourced fold conditions, we fix fold 2 as the test split and use folds 1, 3, 4, and 5 as training folds. Unless otherwise specified, both propagation passes use Topology-DAES, the model branch is warmed up for 20 epochs, and the model-evidence cap is $w_{\max} = 0.5$. Table 2 summarizes the fixed hyperparameters used in the main experiments.

4.2. Main Results

Table 3 shows that NSE is consistently competitive on CIFAR-10 and CIFAR-100. The strongest gains appear on CIFAR-100, where the native candidate source is sparse and true-label absence is more damaging; at $q = 0.05, \eta = 0.3$, NSE reaches $79.22\% \pm 0.14$, improving over PALS by 1.35 points.

Table 4 evaluates higher noise rates, where many samples miss their true label. NSE remains ahead of PALS and ALIM, suggesting that active extraction from the restored prior is still effective even when source membership is frequently incomplete.

4.3. Component Ablations

Table 8 isolates the implemented components on CIFAR-100 with $q = 0.05, \eta = 0.3$. All rows use 500 epochs and three seeds. The baseline row matches the main NSE protocol used in Table 3.

The largest drop comes from replacing the first Topology-DAES pass, either alone or together with the second pass. This supports the claim that robust extraction starts from topology-controlled propagation of the restored prior. Removing the candidate-prior projection from model evidence costs about one point, indicating that the native source is still useful as a constraint even under NPLL noise. The confidence-ratio gate and model cap have smaller effects; they are stabilizing choices rather than the main source of performance. Disabling salvage reduces accuracy by 0.55 points, supporting the role of consensus recovery for samples outside the conservative reliable set.

4.4. Persistent Supervision State Proxies

The core claim of NSE is not only that its extraction modules work, but that NPLL does not require inheriting model-induced supervision state across epochs. We therefore run a controlled persistent-state proxy study on CIFAR-100 ($q = 0.05, \eta = 0.3$). These rows are not full reimplementations of the prior methods. Instead, each row transplants the core inherited-supervision mechanism of a representative method into the same NSE extractor, model, warm-up schedule, active-set training, and salvage logic. This isolates the effect of carrying candidate membership, soft pseudo-targets, or reliable-promotion state into later epochs. The proxies are intentionally conservative and mechanism-aligned. UPLLRS-PSS uses the NSE-estimated noisy set because UPLLRS is built around a reliable/unreliable split; promoted labels are selected only from the native non-candidate space. IRNet-PSS does not use the NSE-estimated noisy set, because IRNet defines its own noisy-sample detector through the gap between the best candidate and best non-candidate scores. FREDIS-PSS follows the score-gap refinement/disambiguation rule and additionally requires high-confidence non-candidate insertion and low-confidence candidate removal to avoid an overly aggressive proxy. PALS/SARI-PSS keeps the all-sample confident top-1 partial-label augmentation rule, but uses a high threshold schedule ($0.95 \rightarrow 0.85$) because SARI's original lower threshold is coupled with label smoothing and its full training pipeline. PiCO+-PSS isolates persistent soft pseudo-target carry-over; when prototype scores are not reproduced, the proxy uses the second-pass NSE evidence $P^{(2)}$ as the soft target evidence. We keep Peng et al.'s candidate-label reconstruction framework in Related Work but exclude it from this controlled proxy table because it couples persistent candidate reconstruction with a separate feature-space reconstruction pipeline; the table below tests only supervision-state carry-over under the same NSE extractor.

Tables 9 and 10 directly test the restoration claim. NSE inherits model parameters, prototypes, and diagnostic histories, but it does not inherit model-induced supervision state. The proxy rows test whether carrying the corresponding prior-work mechanism into later epochs improves the same extractor or instead introduces wrong writes, wrong

Parameter	Value	Role
K	15	nearest neighbors per sample
δ	0.25	class-balanced reliable-selection quantile
γ	2.0	masked-entropy reliability strength
τ_{sp}	0.5	spatial softmax temperature
τ_0	0.1	base DAES propagation temperature
λ_H	0.5	entropy-adaptive temperature coefficient
p_{sim}	2	squared similarity power
T_{warm}	20	model-evidence warm-up epochs
$w_{\max}$	0.5	maximum model-evidence contribution
L_{hist}	15	salvage history length
α_{mix}	1.0	MixUp Beta parameter
ϵ_{ls}	0.0	label smoothing value
β	0.9	prototype EMA coefficient

Table 2. Default hyperparameters for NSE. Values are fixed across the main CIFAR experiments unless an ablation explicitly changes them.

CIFAR-10	$q = 0.1$			$q = 0.3$			$q = 0.5$		
	$\eta = 0.1$	$\eta = 0.2$	$\eta = 0.3$	$\eta = 0.1$	$\eta = 0.2$	$\eta = 0.3$	$\eta = 0.1$	$\eta = 0.2$	$\eta = 0.3$
RC	80.87±0.30	78.22±0.23	75.24±0.17	79.69±0.37	75.69±0.63	71.01±0.54	72.46±1.51	59.72±0.42	49.74±0.70
LWS	82.97±0.24	79.46±0.09	74.28±0.79	80.93±0.28	76.07±0.38	69.70±0.72	70.41±2.68	58.26±0.28	39.42±3.09
FREDIS	90.57±0.23	88.01± 0.19	84.35±0.20	89.02±0.15	86.14±0.17	81.02±0.60	87.42±0.21	80.81±0.99	65.15±0.13
PiCO	90.78±0.24	87.27±0.11	84.96±0.12	89.71±0.18	85.78±0.23	82.25±0.32	88.11±0.29	82.41±0.30	68.75±2.62
CRDPLL	93.48±0.17	89.13±0.39	86.19±0.48	92.73±0.19	86.96±0.21	83.40±0.14	91.10±0.07	82.30±0.46	73.78±0.55
IRNet	93.44±0.21	92.57±0.25	92.38±0.21	92.81±0.19	92.18±0.18	91.35±0.08	91.51±0.05	90.76±0.10	86.19±0.41
PiCO+	94.48±0.02	94.74±0.13	94.43±0.19	94.02±0.03	94.03±0.01	92.94±0.24	93.56±0.08	92.65±0.26	88.21±0.37
UPLLRS	95.16±0.10	-	94.65±0.23	94.32±0.21	-	93.85±0.31	92.47±0.19	-	91.55±0.38
ALIM-Scale	95.71±0.01	95.50±0.08	95.35±0.13	95.31±0.16	94.77±0.07	94.36±0.03	94.71±0.04	93.82±0.13	90.63±0.10
ALIM-Onehot	95.83±0.13	95.86±0.15	95.75±0.19	95.52±0.15	95.41±0.13	94.67±0.21	95.19±0.24	93.89±0.21	92.26±0.29
PALS	96.28±0.05	96.17±0.18	95.90±0.20	95.96±0.08	95.76±0.12	95.43±0.17	95.52±0.13	**95.89±0.14**	94.18±0.10
NSE	**96.72±0.11**	**96.58±0.05**	**96.71±0.07**	**96.33±0.08**	**96.19±0.12**	**96.03±0.08**	**95.93±0.19**	95.19±0.13	**94.82±0.04**

CIFAR-100	$q = 0.01$			$q = 0.03$			$q = 0.05$		
	$\eta = 0.1$	$\eta = 0.2$	$\eta = 0.3$	$\eta = 0.1$	$\eta = 0.2$	$\eta = 0.3$	$\eta = 0.1$	$\eta = 0.2$	$\eta = 0.3$
RC	52.73±1.05	48.59±1.04	45.77±0.31	52.15±0.19	48.25±0.38	43.92±0.37	46.62±0.34	45.46±0.21	40.31±0.55
LWS	56.05±0.20	50.66±0.59	45.71±0.45	53.59±0.45	48.28±0.44	42.20±0.49	45.46±0.44	39.63±0.80	33.60±0.64
FREDIS	64.73±0.28	64.53 ± 0.39	59.42±0.18	67.38±0.40	63.86±0.50	60.86±0.72	66.43±0.22	63.09 ± 0.22	56.15±0.18
PiCO	68.27±0.08	62.24±0.31	58.97±0.09	67.38±0.09	62.01±0.33	58.64±0.28	67.52±0.43	61.52±0.28	58.18±0.65
CRDPLL	68.12±0.13	65.32±0.34	62.94±0.28	67.53±0.07	64.29±0.27	61.79±0.11	67.17±0.04	64.11±0.42	61.03±0.43
IRNet	71.17±0.14	70.10±0.28	68.77±0.28	71.01±0.43	70.15±0.17	68.18±0.30	70.73±0.09	69.33±0.51	68.09±0.12
PiCO+	75.04±0.18	74.31±0.02	71.79±0.17	74.68±0.19	73.65±0.23	69.97±0.01	73.06±0.16	71.37±0.16	67.56±0.17
UPLLRS	75.73±0.41	-	71.72±0.39	-	-	-	74.73±0.24	-	70.31±0.22
ALIM-Scale	77.37±0.32	76.81±0.05	76.45±0.30	77.60±0.18	76.63±0.19	75.92±0.14	76.86±0.23	76.44±0.12	75.67±0.17
ALIM-Onehot	76.52±0.19	76.55±0.24	76.09±0.23	77.27±0.23	76.29±0.41	75.29±0.57	76.87±0.20	75.23±0.42	74.49±0.61
PALS	80.90±0.50	80.45±0.49	79.78±0.13	80.33±0.04	79.40±0.50	78.52±0.24	80.00±0.25	79.08±0.22	77.87±0.29
NSE	**81.62±0.09**	**81.34±0.17**	**80.76±0.34**	**81.10±0.11**	**80.44±0.19**	**79.47±0.06**	**80.32±0.47**	**80.05±0.20**	**79.22±0.14**

Table 3. Comparison with previous methods on synthetic NPLL benchmarks. Baseline numbers follow the PALS-reported comparison tables; missing entries are kept unreported when absent from that comparison.

CIFAR-100	$q = 0.05$ $\eta = 0.4$	$q = 0.05$ $\eta = 0.5$
FREDIS	53.44±0.39	48.05±0.20
PiCO	44.17±0.08	35.51±1.14
CRDPLL	57.10±0.24	52.10±0.36
UPLLRS	-	64.78±0.53
PiCO+	66.41±0.58	60.50±0.99
ALIM-Scale	74.98±0.16	72.26±0.25
ALIM-Onehot	71.76±0.56	69.59±0.62
PALS	76.72±0.41	74.79±0.40
NSE	**78.04±0.06**	**76.69±0.46**

Table 4. Extreme-noise results on CIFAR-100.

CIFAR-100	$q = 0.01$	$q = 0.05$	$q = 0.1$
RC	75.36±0.06	74.44±0.31	73.79±0.29
LWS	64.55±1.98	50.19±0.34	44.93±1.09
PiCO	73.78±0.15	72.78±0.38	71.55±0.31
CRDPLL	79.74±0.07	78.97±0.13	78.51±0.24
PiCO+	76.29±0.42	76.17±0.18	75.55±0.21
PALS	81.46±0.13	81.00±0.26	80.77±0.12
NSE	**82.25±0.12**	**81.27±0.16**	**80.79±0.20**

Table 5. Noise-free PLL results on CIFAR-100.

promotions, source drift, soft-mass weakening, or clean-sample damage. All five inherited-state proxies underperform NSE-Reset in final accuracy: PALS/SARI-PSS is the least harmful but still drops by 0.54 points, FREDIS-PSS and IRNet-PSS drop by 1.28 and 1.34 points, UPLLRS-PSS drops by 0.87 points, and the soft PiCO+-style carry-over drops by 19.70 points. The soft carry-over row is especially diagnostic: it recovers the true label somewhere in the soft support for all initially noisy samples ($\mathrm{SrcRec\text{-}N} = 1.000$), but it also writes every row at every epoch, has about 40%

Dataset	CIFAR-100H
q	0.5
η	0.2
PiCO	59.8±0.25
PiCO+	68.31±0.47
ALIM-Scale	73.42±0.18
ALIM-Onehot	72.36±0.20
PALS	75.16±0.59
NSE	**76.54±0.18**

Table 6. Hierarchical CIFAR-100H result.

	Treeversity	Benthic	Plankton
Divide-Mix	77.84±0.52	71.72±1.21	91.79±0.51
ELR+	79.05±1.22	67.53±1.76	91.76±0.91
PiCO	**84.74±0.42**	51.01±0.74	24.95±0.13
PALS	81.56±0.28	**77.50±0.35**	90.04±0.50
NSE	83.76±0.50	77.24±0.32	**92.68±0.14**

(a) Ten human annotators

	Treeversity	Benthic	Plankton
Divide-Mix	76.38±1.78	71.50±0.42	91.96±0.52
ELR+	77.07±0.43	69.35±0.97	91.36±0.40
ALIM-Onehot	82.58±0.92	49.17±3.06	23.75±2.53
PALS	80.47±1.22	76.52±0.03	90.20±0.97
NSE	**83.77±0.97**	**76.82±0.42**	**92.76±0.14**

(b) Three human annotators

Table 7. Crowdsourced datasets. NSE is strongest on Plankton and remains competitive on Treeversity and Benthic, where margins are within or close to the standard deviations of the strongest baselines.

wrong top-1 write events, and accumulates large source drift. This shows why source-level repair is not equivalent to reliable active supervision. NSE instead separates source-level recovery from active-set recovery: it does not try to write missing true labels back into the native source record, but recovers them through active supervision while preserving the native source record.

5. Conclusion

In this paper, we introduced NSE, a Non-Destructive Supervision Extraction framework for Noisy Partial Label Learning (NPLL). NSE shows that high-performance NPLL can be achieved without cross-epoch persistent source write-back. The key is not to leave weak supervision unprocessed, but to restore the native candidate/crowd prior at the beginning of each epoch and extract reliable active supervision from the restored working prior.

Topology-DAES is the core extraction module of NSE. By combining masked-entropy topology reliability with entropy-adaptive KNN propagation, it suppresses ambiguous neighborhoods and strengthens consistent weak-label neighborhoods. Prior-constrained model evidence and active-set training are then built on top of this topology-aware geometric belief. These results suggest that the future direction of NPLL should not be limited to candidate-set purification, reconstruction, or persistent pseudo-target carry-over. A complementary and effective direction is to decouple the native weak-supervision record from the epoch-local working supervision and improve how temporary active supervision is extracted from it.

NSE still depends on the quality of the learned representation and the semantic purity of local neighborhoods. When neighborhoods are strongly mixed or the crowd/candidate prior provides little recoverable structure, epoch-wise restoration alone cannot guarantee reliable extraction. This limitation makes active-set precision, coverage, and source-drift diagnostics important parts of evaluating future NPLL methods.

ID	Ablation	Final Acc.	Δ
A0	full NSE: two-pass Topology-DAES, adaptive r_i, candidate-prior-projected model evidence, active salvage	79.22±0.14	–
A1	replace both Topology-DAES passes with a fixed exponential kernel	76.50±0.20	-2.72
A2	replace only the second Topology-DAES pass with a fixed exponential kernel	78.56±0.05	-0.66
A3	replace only the first Topology-DAES pass with a fixed exponential kernel	76.51±0.12	-2.71
A4	use constant $r_i = 0.5$ instead of confidence-ratio arbitration	79.14±0.11	-0.08
A5	remove the candidate-prior projection from model evidence	78.17±0.06	-1.05
A6	set $w_{\max} = 0.0$ so the second-stage input is KNN-only	78.99±0.04	-0.23
A7	set $w_{\max} = 1.0$ after warm-up	79.13±0.13	-0.09
A8	detect salvage candidates but do not promote them into active training	78.67±0.29	-0.55

Table 8. Component ablations on CIFAR-100 ($q = 0.05, \eta = 0.3$).

Condition	Prior-aligned inherited state	Proxy operator	Host	Best Acc.	Final Acc.	Prec(A)	Cov(A)	NRR-A
NSE-Reset	none; restore $\tilde{\Omega}^t \leftarrow \Omega^0$ each epoch	none	c201	79.55±0.04	79.38±0.16	0.932	0.887	0.660
FREDIS-V2-PSS	refinement plus disambiguation candidate state	add top non-candidate if score gap ≤ 0.05 and confidence ≥ 0.85; remove candidates with gap ≥ 0.85 and confidence ≤ 0.05	c201	78.42±0.17	78.10±0.20	0.916	0.873	0.640
IRNet-V2-PSS	noisy-sample correction state	insert best non-candidate if it exceeds the best candidate and its confidence is at least 0.85; no swapping	c201	78.26±0.34	78.04±0.27	0.917	0.873	0.641
PALS/SARI-V2-PSS	partial-label augmentation state	all-sample top-1 augmentation with a conservative linear threshold $0.95 \to 0.85$	c201	79.00±0.23	78.84±0.32	0.925	0.881	0.647
UPLLRS-V2-PSS	persistent reliable-promotion state	from NSE-estimated noisy samples, store high-confidence native non-candidate pseudo-labels and force them into later active training	c201	78.68±0.10	78.51±0.12	0.921	0.884	0.647
PiCO+-V2-PSS	persistent soft pseudo-target state	blend the next epoch soft target with $P^{(2)}$ using $\alpha_{\mathrm{wb}} = 0.1$	c201	59.90±0.67	59.68±0.65	0.606	0.975	0.589

Table 9. Persistent supervision-state proxy study on CIFAR-100 ($q = 0.05, \eta = 0.3$). All proxy rows keep the NSE extractor fixed and change only the inherited supervision state. The proxy rows are controlled diagnostics, not complete reproductions of the cited methods.

Condition	Write/Promote	Wrong	SrcRec-N	MassRec-N	Damage/Mass	Drift	AUC-Drift
NSE-Reset	0.000	–	0.000	0.000	0.000	0.000	0.000
FREDIS-V2-PSS	0.000	0.000	0.182	0.177	0.002	0.394	0.308
IRNet-V2-PSS	0.000	0.000	0.173	0.086	0.001	0.065	0.050
PALS/SARI-V2-PSS	0.000	0.000	0.105	0.053	0.001	0.037	0.022
UPLLRS-V2-PSS	0.029	0.334	0.000	0.000	0.000	0.000	0.000
PiCO+-V2-PSS	1.000	0.402	1.000	0.152	0.107	1.787	1.687

Table 10. Final-epoch persistent-state diagnostics for Table 9. Write is used for candidate or soft source-state updates; Promote is used for UPLLRS-style sample-state promotion. For UPLLRS-PSS, Wrong reports cumulative wrong promotion among promoted samples; for other rows, Wrong is the final-epoch wrong-write rate when final-epoch writes occur. Soft states are additionally evaluated by MassRec-N and DamageMass.

References

[1] Timothee Cour, Benjamin Sapp, Chris Jordan, and Ben Taskar. Learning from ambiguously labeled images. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pages 919–926. IEEE, 2009. 1, 2

[2] Lei Feng, Jiaqi Lv, Bo Han, Miao Xu, Gang Niu, Xin Geng, Bo An, and Masashi Sugiyama. Provably consistent partial-label learning. *Advances in neural information processing systems*, 33:10948–10960, 2020. 1, 3, 9

[3] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016. 10

[4] Alex Krizhevsky and Geoffrey Hinton. Learning multiple layers of features from tiny images. Technical report, University of Toronto, Toronto, Ontario, 2009. 9

[5] Zheng Lian, Mingyu Xu, Lan Chen, Licai Sun, Bin Liu, Lei Feng, and Jianhua Tao. IRNet: Iterative Refinement Network for Noisy Partial Label Learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 48(2):1932–1948, 2026. 1, 3, 4, 7, 9

[6] Jiaqi Lv, Miao Xu, Lei Feng, Gang Niu, Xin Geng, and Masashi Sugiyama. Progressive identification of true labels for partial-label learning. In *International conference on machine learning*, pages 6500–6510. PMLR, 2020. 3, 4

[7] Xiaorui Peng, Yuheng Jia, Fuchao Yang, Ran Wang, and Min-Ling Zhang. Noise separation guided candidate label reconstruction for noisy partial label learning. In *International Conference on Learning Representations*, 2025. 1, 3, 4, 7

[8] Congyu Qiao, Ning Xu, and Xin Geng. Decompositional generation process for instance-dependent partial label learning. In *International Conference on Learning Representations*, 2023. 1, 3

[9] Congyu Qiao, Ning Xu, Jiaqi Lv, Yi Ren, and Xin Geng. Fredis: A fusion framework of refinement and disambiguation for unreliable partial label learning. In *International Conference on Machine Learning*, pages 28321–28336. PMLR, 2023. 1, 3, 4, 7, 9

[10] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, et al. Imagenet large scale visual recognition challenge. *International journal of computer vision*, 115:211–252, 2015. 10

[11] Darshana Saravanan, Naresh Manwani, and Vineet Gandhi. Pseudo-labelling meets label smoothing for noisy partial label learning. In *CVPR Workshops*, pages 2152–2161, 2025. 1, 3, 4, 9

[12] Lars Schmarje, Vasco Grossmann, Claudius Zelenka, Sabine Dippel, Rainer Kiko, Mariusz Oszust, Matti Pastell, Jenny Stracke, Anna Valros, Nina Volkmann, et al. Is one annotation enough?-a data-centric image classification benchmark for noisy and ambiguous label estimation. *Advances in Neural Information Processing Systems*, 35:33215–33232, 2022. 9

[13] Yu Shi, Ning Xu, Hua Yuan, and Xin Geng. Unreliable partial label learning with recursive separation. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence*, 2023. 1, 3, 4, 7, 9

[14] Shiyu Tian, Hongxin Wei, Yiqun Wang, and Lei Feng. CroSel: Cross selection of confident pseudo labels for partial-label learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 19479–19488, 2024. 3, 4, 8

[15] Haobo Wang, Ruixuan Xiao, Yixuan Li, Lei Feng, Gang Niu, Gang Chen, and Junbo Zhao. PiCO: Contrastive label disambiguation for partial label learning. In *International Conference on Learning Representations*, 2022. 1, 3, 4, 7, 9

[16] Haobo Wang, Ruixuan Xiao, Yixuan Li, Lei Feng, Gang Niu, Gang Chen, and Junbo Zhao. Pico+: Contrastive label disambiguation for robust partial label learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(5): 3183–3198, 2024. 1, 3, 4, 7, 9

[17] Hongwei Wen, Jingyi Cui, Hanyuan Hang, Jiabin Liu, Yisen Wang, and Zhouchen Lin. Leveraged weighted loss for partial label learning. In *International Conference on Machine Learning*, pages 11091–11100. PMLR, 2021. 3, 4, 9

[18] Dong-Dong Wu, Deng-Bao Wang, and Min-Ling Zhang. Revisiting consistency regularization for deep partial label learning. In *International Conference on Machine Learning*, pages 24212–24225. PMLR, 2022. 3, 9

[19] Mingyu Xu, Zheng Lian, Lei Feng, Bin Liu, and Jianhua Tao. Alim: adjusting label importance mechanism for noisy partial label learning. In *Proceedings of the 37th International Conference on Neural Information Processing Systems*, Red Hook, NY, USA, 2023. Curran Associates Inc. 1, 3, 4, 7, 9

[20] Ning Xu, Congyu Qiao, Xin Geng, and Min-Ling Zhang. Instance-dependent partial label learning. In *Advances in Neural Information Processing Systems*, pages 27119–27130, 2021. 1, 3

[21] Ning Xu, Biao Liu, Jiaqi Lv, Congyu Qiao, and Xin Geng. Progressive purification for instance-dependent partial label learning. In *Proceedings of the 40th International Conference on Machine Learning*, pages 38551–38565. PMLR, 2023. 1, 3, 4, 7

[22] Fei Yu and Min-Ling Zhang. Maximum margin partial label learning. In *Asian conference on machine learning*, pages 96–111. PMLR, 2016. 3

[23] Min-Ling Zhang and Fei Yu. Solving the partial label learning problem: An instance-based approach. In *IJCAI*, pages 4048–4054, 2015. 1, 3